

姓名:

学院:

学号:

上海科技大学

2023-2024 学年第 2 学期本科生期中考试卷

开课单位: 信息学院

授课教师: 邵子瑜

考试科目: 《面向信息科学的概率论与数理统计》

课程代码: SI 140A

考试时间: 2024 年 4 月 16 日 10 点 15 分 - 11 点 55 分。

考试成绩录入表:

题目	1	2	3	4	5	6	7	8	总分
计分									
复核									

评卷人签名:

日期:

复核人签名:

日期:

编写说明:

1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定, 每页须按顺序标注页码(除封面外), 要求排版清晰、美观, 便于在页面左侧装订。为方便印刷归档, 建议使用 A4 双面印刷(学校有印刷一体机提供)。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理, 如考试过程中发现试题有误, 主考教师需负责现场解释, 此类情况学校将作为教学评估记录的一部分。

Probability & Statistics for EECS
Spring 2024
Midterm
2024/04/16
Time Limit: 100 Minutes

Name (Print): _____

Advisor Name _____

This exam contains 10 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

Try to answer as many problems as you can. The following rules apply:

- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.

Do not write in the table to the right.

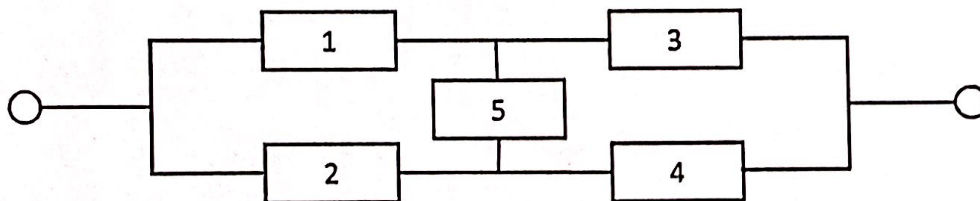
Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	20	
8	20	
Total:	100	

1. (10 points) How many number of distinct positive integer-valued vectors $(x_1, x_2, \dots, x_5)$ satisfying the equation and inequalities

$$x_1 + x_2 + \dots + x_5 = 48$$

$$x_1 > 3, x_2 \geq 7, x_3 > 4, x_4 \geq 6, x_5 > 4.$$

2. (10 points) A system composed of 5 homogeneous devices is shown in the following figure. It is said to be functional when there exists at least one end-to-end path that devices on such path are all functional. For such a system, if each device, which is independent of all other devices, functions with probability $p = 1/2$, then what is the probability that the system functions? Such a probability is also called the system reliability.



3. (10 points) A frog starts at 0 on the number line and makes a sequence of jumps to the right. In any one jump, independent of previous jumps, this frog leaps a positive integer distance m with probability $\frac{1}{2^m}$. Find the probability that this frog will eventually land at 1997 on the number line.

4. (10 points) A six-sided fair dice is rolled four times independently. What is more likely: a sum of 15 or a sum of 16? (You need to compute the corresponding probability for each case).

5. (10 points) Consider a coin that comes up heads with probability p and tails with probability $1-p$. Let q_n be the probability that after n independent tosses, there have been an odd number of heads. Find q_n .

6. (10 points) An urn contains w white balls and b black balls, which are randomly drawn one by one without replacement. Let X denote the number of black balls drawn before drawing r ($1 \leq r \leq w$) white balls.
- (a) (5 points) Find the PMF of X .
 - (b) (5 points) Find $E(X)$.

7. (20 points) Bob's database of friends contains n entries, but due to a software bug, the addresses correspond to the names in a totally random fashion. Bob writes a holiday card to each of his friends and sends it to the (software-corrupted) address. Let X denote the number of friends of him who will get the correct card.
- (a) (5 points) Find $E(X)$.
 - (b) (5 points) Find $\text{Var}(X)$.
 - (c) (5 points) Find the PMF of X .
 - (d) (5 points) When $n \rightarrow \infty$, show that the distribution of X converges to a Poisson distribution.

8. (20 points) Consider the following 7-door version of the Monty Hall problem. There are 7 doors, behind one of which there is a car (which you want), and behind the rest of which there are goats (which you don't want). Initially, all possibilities are equally likely for where the car is. You choose a door. Monty Hall then opens 3 goat doors, and offers you the option of switching to any of the remaining 3 doors.
- (a) (10 points) Assume that Monty Hall knows which door has the car, will always open 3 goat doors and offer the option of switching, and that Monty chooses with equal probabilities from all his choices of which goat doors to open. Should you switch? What is your probability of success if you switch to one of the remaining 3 doors?
 - (b) (10 points) Generalize the above to a Monty Hall problem where there are $n \geq 3$ doors, of which Monty opens m goat doors, with $1 \leq m \leq n - 2$.

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

	Y discrete	Y continuous
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y)dy$
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y)dy$

Figure 1: Bayes' Rule & LOTP.

Table of distributions

Name	Param.	PMF or PDF	Mean	Variance
Bernoulli	p	$P(X = 1) = p, P(X = 0) = q$	p	pq
Binomial	n, p	$\binom{n}{k} p^k q^{n-k}$, for $k \in \{0, 1, \dots, n\}$	np	npq
FS	p	pq^{k-1} , for $k \in \{1, 2, \dots\}$	$1/p$	q/p^2
Geom	p	pq^k , for $k \in \{0, 1, 2, \dots\}$	q/p	q/p^2
NBinom	r, p	$\binom{r+n-1}{r-1} p^r q^n$, $n \in \{0, 1, 2, \dots\}$	$r q/p$	$r q/p^2$
HGeom	w, b, n	$\frac{\binom{n}{k} \binom{w+b-k}{n-k}}{\binom{w+b}{n}}$, for $k \in \{0, 1, \dots, n\}$	$\mu = \frac{nw}{w+b}$	$(\frac{w+b-n}{w+b-1}) n \frac{w}{n} (1 - \frac{w}{n})$
Poisson	λ	$\frac{e^{-\lambda} \lambda^k}{k!}$, for $k \in \{0, 1, 2, \dots\}$	λ	λ
Uniform	$a < b$	$\frac{1}{b-a}$, for $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Log-Normal	μ, σ^2	$\frac{1}{x\sigma\sqrt{2\pi}} e^{-(\ln x - \mu)^2/(2\sigma^2)}$, $x > 0$	$\theta = e^{\mu + \sigma^2/2}$	$\theta^2(e^{\sigma^2} - 1)$
Expo	λ	$\lambda e^{-\lambda x}$, for $x > 0$	$1/\lambda$	$1/\lambda^2$
Gamma	a, λ	$\Gamma(a)^{-1} (\lambda x)^a e^{-\lambda x} x^{a-1}$, for $x > 0$	a/λ	a/λ^2
Beta	a, b	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$, for $0 < x < 1$	$\mu = \frac{a}{a+b}$	$\frac{\mu(1-\mu)}{a+b+1}$
Chi-Square	n	$\frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}$, for $x > 0$	n	$2n$
Student-t	n	$\frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} (1 + x^2/n)^{-(n+1)/2}$	0 if $n > 1$	$\frac{n}{n-2}$ if $n > 2$